A Seminar Report

On

"Smart Tree Adoption System"

Submitted to

Savitribai Phule PuneUniversity, Pune

Inpartial fulfilment for the award of the Degree
of Bachelor of Engineering in

Information Technology

By

Adhiraj Sikandar Kate (I3273)

Under the guidance of

Ms A.S Thorat

Sinhgad Institutes

Department of Information Technology STES's, Smt. Kashibai Navale College of Engineering,

Vadgaon (BK), Pune, 411 041.

2026-27 (SEM-I)

Sinhgad Institutes

CERTIFICATE

This is to certify that the seminar report entitled "Smart Tree Adoption System: An AIEnabled Platform for Sustainable Environmental Conservation" being submitted by Adhiraj Sikandar Kate record of bonafide work carried out by him under the supervision and guidance of prof. A.S Thorat in partial fulfillment of the requirement for **TE (InformationTechnology)**ˆ 2024 course of

Savitribai Phule Pune University,Pune in the academic year 2026-27

Date:24 /09 /2026

Place: Pune

Prof A.S Thorat
Guide

Dr. M. L. Bangare
Head of the Department, IT

Dr. A. V. Deshpande
Principal, SKNCOE, Pune

This Seminar report has been examined by Savitribai Phule Pune University,Pune requirements at **Smt. Kashibai Navale College of Engineering, Pune-41** on

_____

InternalExaminer

ExternalExaminer

“Smart Tree Adoption System

: An AI-Enabled Platform for Sustainable Environmental Conservation”

ACKNOWLEDGEMENT

Purpose of acknowledgements page is to show appreciation to those who contributed in conducting this seminar work/ other tasks and duties related to the report writing. Therefore when writing acknowledgements page you should carefully consider everyone who helped during research process and show appreciation in the order of relevance. In this regard it is suitable to show appreciation in brief manner instead of using strong emotional phrases.

In this part of your work it is normal to use personal pronouns like ”I, my, me” while in the rest of the report this articulation is not recommended. Even when acknowledging family members and friends make sure of using the wording of a relatively formal register. The list of the persons you should acknowledged, includes guide (main and second), academic staff in your de- partment, technical staff, reviewers, companies, family and friends.

You should acknowledge all sources of funding. Its usually specific nam- ing the person and the type of help you received. For example, an advisor who helped you conceptualize the project, someone who helped with the actual building or procedures used to complete the project, someone who helped with computer knowledge, someone who provided raw materials for the project, etc.

Adhiraj Sikandar Kate

TABLE OF CONTENTS

1. Abstract and Keywords.
2. Introduction with section including organization of the report.
3. Literature Survey.
4. Motivation, purpose and scope and objective of seminar.
5. Details of design.
6. Conclusions.
7. Bibliography/References (in IEEE Format).
8. Copyright Work. 9. Plagiarism Check report.

ABSTRACT

The Smart Tree Adoption System: An AI-Enabled Platform for Sustainable Environmental Conservation is a technology-driven platform designed to encourage tree adoption and promote sustainable environmental conservation. The system connects individuals, organizations, and communities with tree-planting initiatives, allowing users to adopt trees, monitor their growth, and track their environmental contributions. Artificial Intelligence can be used to analyze tree health, identify potential diseases, estimate growth, and provide personalized recommendations for maintenance. The platform can also maintain digital records of adopted trees, locations, adoption history, and environmental impact such as estimated carbon absorption. By combining AI, digital monitoring, and community participation, the proposed system aims to improve tree survival, increase awareness about environmental sustainability, and encourage long-term participation in conservation activities.

Keywords: Agentic Artificial Intelligence (Agentic AI); Large Language Models (LLMs); Hybrid Retrieval-Augmented Generation (RAG); Knowledge Base.

LIST OF FIGURES

Figure No.	Figure Name	Page No.
1.1	Smart Tree Adoption System Overview	8
1.2	Proposed System Architecture	9
1.3	Tree Adoption Workflow	10
1.4	AI-Based Tree Health Monitoring	11
2.1	IoT-Based Tree Health Monitoring	13
2.2	AI-Based Tree Health Assessment	14
2.3	UAV-Based Tree Monitoring	16
2.4	Comparison of Existing and Proposed Systems	18
3.1	Methodology of Smart Tree Adoption System	23
3.2	System Architecture and Data Flow	24
3.3	Future Enhancements of Proposed System	26

LIST OF TABLES

Table No.	Table Name	Page No.
1.1	Technologies Used in Smart Tree Adoption System	12
2.1	Summary of Related Research Work	19
2.2	Comparison of Existing Approaches	20
2.3	Advantages and Disadvantages of Existing Systems	21
3.1	Advantages and Disadvantages of Proposed System	25
3.2	Applications of Smart Tree Adoption System	26

ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
IoT	Internet of Things
CNN	Convolutional Neural Network
UAV	Unmanned Aerial Vehicle
GPS	Global Positioning System
GIS	Geographic Information System
CSR	Corporate Social Responsibility
CNN	Convolutional Neural Network
U-Net	U-Net Neural Network Architecture
LoRa	Long Range
YOLO	You Only Look Once
DEM	Digital Elevation Model
DTM	Digital Terrain Model
WebGIS	Web Geographic Information System

Chapter 1 INTRODUCTION

1. INTRODUCTION

1.1 Introduction to Domain

Environmental conservation is an important domain focused on protecting natural resources, improving ecological balance, and reducing the harmful effects of human activities on the environment. Rapid urbanization, deforestation, pollution, and climate change have increased the need for effective and sustainable environmental management. Tree plantation and conservation play a major role in improving air quality, reducing carbon dioxide levels, preventing soil erosion, and supporting biodiversity. With the advancement of information technology, Artificial Intelligence (AI), Internet of Things (IoT), cloud computing, and digital platforms can be used to improve environmental monitoring and conservation activities. These technologies provide opportunities to monitor trees, analyze their health, maintain digital records, and encourage greater participation from individuals and organizations in environmental initiatives.

1.2 Introduction to Seminar Topic

The Smart Tree Adoption System: An AI-Enabled Platform for Sustainable Environmental Conservation is a technology-based platform designed to encourage individuals and organizations to adopt and care for trees. The system allows users to select and adopt trees, maintain their adoption records, monitor tree growth, and receive information about their environmental contribution. AI can be integrated into the system to analyze tree images, identify possible health problems, and provide recommendations for tree maintenance. The platform can also maintain information such as tree location, species, adoption date, growth status, and estimated environmental impact. By combining artificial intelligence with digital tree monitoring and community participation, the proposed system aims to promote responsible tree care, increase environmental awareness, and support long-term sustainable conservation efforts.

•

CHAPTER 2 BACKGROUND WORK OF SMART TREE ADOPTION SYSTEM

2.1 Introduction

The increasing use of Artificial Intelligence, Internet of Things (IoT), remote sensing, machine learning, and web-based platforms has created new possibilities for environmental conservation and tree management. Researchers have developed different systems for monitoring tree health, detecting diseases, mapping trees, measuring environmental conditions, and encouraging public participation in plantation activities. This chapter reviews existing research and technologies related to tree monitoring, AI-based health assessment, IoT-based monitoring, digital conservation platforms, and tree adoption systems. The advantages and limitations of these approaches are discussed to identify the research gap addressed by the proposed **Smart Tree Adoption System**.

2.2 Existing Related Work

2.2.1 IoT-Based Tree Health Monitoring

An IEEE study, **“IoT-based System for Monitoring Health State of Trees,”** proposed an IoT architecture in which sensors installed around trees collect health-related information and communicate with a central gateway using LoRa technology. Such systems can provide continuous monitoring without depending entirely on manual inspection.

Advantages:

- Continuous monitoring of tree conditions.
- Automated collection of environmental data.
- Wireless communication using LoRa.
- Reduces the need for frequent manual inspection.

Disadvantages:

- Requires physical sensors and hardware installation.
- Maintenance and replacement of sensors can increase cost.
- Sensor deployment at a large scale can be difficult.
- Does not primarily focus on user-driven tree adoption.

2.2.2 AI and Deep Learning for Tree Health Assessment

Recent research has investigated deep learning techniques for automatically assessing tree health from images. A 2026 study developed a CNN-based semantic segmentation framework using **U-Net and ResNet** models to identify foliage, wood, ivy, and background and derive indicators such as defoliation, tree height, crown structure, and tree risk.

Advantages:

- Automated analysis of tree images.
- Can identify visible health indicators.

-
- Reduces dependence on manual assessment.

  Can support large-scale tree monitoring.

Disadvantages:

- Requires suitable training datasets.
- Accuracy can be affected by image quality and visibility.
- Deep learning models can require significant computational resources.
- Health assessment alone does not provide a complete tree-adoption and user-engagement platform.

2.2.3 Mobile and Multispectral Tree Monitoring

GreenScan, published in the *IEEE Sensors Journal*, uses mobile ground-based sensing with thermal and multispectral imaging to estimate tree-health indexes. The system was designed to provide treehealth assessments at high spatial and temporal resolution while reducing the cost and labor associated with large-scale inspection.

Advantages:

- Uses multiple types of image information.
- Supports large-scale monitoring.
- Provides automated health assessment.
- Reduces manual inspection requirements.

Disadvantages:

- Requires specialized imaging sensors.
- Implementation can be technically complex.
- Processing multispectral and thermal data requires additional resources.
- Primarily focuses on monitoring rather than tree adoption and community participation.

-

2.2.4 UAV-Based Tree Monitoring

A 2026 study proposed an automated UAV-based system using **YOLOv8, image processing, DEM/DTM data, and WebGIS** for detecting and characterizing urban trees. The system can estimate tree attributes and assess vegetation health, while WebGIS provides spatial visualization. The study reported different detection performance under sparse and dense canopy conditions.

Advantages:

- Covers large geographical areas.
- Enables automated tree detection.
- Provides spatial information through WebGIS.
- Useful for urban forest management.

Disadvantages:

Requires drones and associated equipment.

- Weather and flying restrictions can affect data collection.
- Processing UAV imagery can be computationally demanding.
- May not be practical for individual citizens adopting trees.

2.2.5 Digital Platforms for Environmental Conservation

Digital platforms have also been investigated as a way to increase public participation in environmental conservation. Research on digital conservation platforms highlights the importance of combining environmental data, data analysis, communication, and participatory governance.

Studies of digital conservation platforms have also found that digital tools can support volunteer participation and centralized collection of environmental observations, although user participation and coordination can affect the effectiveness of such systems.

Advantages:

- Encourages public participation.
- Provides centralized environmental information.
- Makes conservation activities easier to track.
- Can connect volunteers and organizations.

Disadvantages:

- Continued user participation can be difficult to maintain.
- Requires proper data management.
- Digital access and technical knowledge may affect participation.
- Many platforms do not provide AI-based individual tree-health analysis.

•

2.2.6 Tree Adoption and Citizen Engagement Platforms

The **ConnecTree** initiative in Pune is an example of a technology-driven platform focused on mapping tree cover and monitoring sapling survival through citizen volunteers. Its objectives include supporting sponsorship, sapling adoption, citizen engagement, plantation management, and monitoring.

Advantages:

- Encourages citizens to participate in plantation activities.
- Supports sapling adoption and sponsorship.
- Enables mapping and monitoring.
- Connects citizens and organizations.

Disadvantages:

- Requires active participation from citizens and sponsors.

  Monitoring quality can depend on submitted data.
- AI-based automated health diagnosis is not the primary focus.
- Further integration of personalized AI recommendations can improve the platform.

•

2.3 Comparison of Existing Approaches

Existing Approach	Main Technology	Major Advantage	Major Limitation
IoT Tree Monitoring	IoT, Sensors, LoRa	Continuous monitoring	Hardware-dependent
AI Tree Health Assessment	CNN, U-Net, ResNet	Automated health analysis	Requires quality datasets
GreenScan	Thermal + Multispectral Imaging	High-resolution health monitoring	Specialized sensors required
UAV Tree Monitoring	UAV, YOLOv8, WebGIS	Large-area monitoring	Drone and processing requirements
Digital Conservation Platforms	Web/Digital Platforms	Public participation	Maintaining engagement can be difficult
ConnecTree	Web Platform, Mapping, Citizen Engagement	Tree adoption and citizen participation	Limited AI-based health analysis

2.4 Research Gap

Existing research demonstrates strong capabilities in **tree-health monitoring, AI-based image analysis, IoT sensing, geographical mapping, and citizen participation**. However, these capabilities are often implemented separately. AI-based systems mainly focus on tree-health detection, while adoption platforms mainly focus on sponsorship, plantation, and citizen engagement. Research on digital conservation platforms also indicates that maintaining long-term participation and reliable environmental data can be challenging.

Therefore, there is scope for an integrated platform that combines **tree adoption, digital tree records, AI-based health monitoring, growth tracking, environmental-impact information, and user participation** within a single system.

2.5 Difference of Proposed System from Existing Systems

The proposed **Smart Tree Adoption System** aims to combine the major useful aspects of existing approaches into one user-oriented platform. Unlike systems that focus only on monitoring or only on adoption, the proposed system can provide a complete lifecycle for an adopted tree—from **adoption and registration to monitoring, health assessment, maintenance recommendations, and environmental-impact tracking**.

CHAPTER 3 CONCLUSION AND OTHER DETAILS

3.1 Methodology

The Smart Tree Adoption System allows users to register, select and adopt trees, and monitor their growth through a digital platform. AI can analyze tree images to identify possible health issues and provide basic maintenance recommendations. Tree details, adoption records, images, and environmental impact are stored in a database.

3.2 Advantages

- Encourages tree adoption and conservation.
- Provides digital tree monitoring.
- Supports AI-based health detection.
- Increases environmental awareness.
- Helps track tree growth and maintenance.

3.3 Disadvantages

- AI accuracy depends on image and training data quality.
- Requires regular user participation.
- Internet and technical infrastructure are required.
- Development and maintenance may involve costs.

3.4 Applications

- Schools and colleges
- NGOs and environmental organizations
- Municipal corporations
- Corporate CSR programs
- Public parks and communities

3.5 Future Enhancements

The system can be enhanced using **IoT sensors, GIS mapping, drone monitoring, advanced AI models, mobile applications, and carbon-impact tracking**.

3.6 Conclusion

The Smart Tree Adoption System combines AI and digital technology to make tree adoption, monitoring, and maintenance easier. It can increase public participation and support sustainable environmental conservation.

REFERENCES

- [Base Paper] Maccacaro et al. (2026). Data-driven tools for optimizing urban tree planting strategies. City and Environment Interactions, Elsevier.
- “ https://www.sciencedirect.com/science/article/pii/S2590252026000905?utm_source=chatgpt.com “
- Zhang, Y. & Ludwig, F. (2026). Towards “Right Tree, Right Place” in urban environments: A systematic review. Urban Forestry & Urban Greening, Elsevier.
- “ https://www.sciencedirect.com/science/article/pii/S1618866726000750?utm_source=chatgpt.com ”
- Yazdi, H. et al. (2026). A 3D target-driven optimisation tool for tree planting location using temporal tree crown geometry development. npj Urban Sustainability, Springer Nature.
- “ https://www.nature.com/articles/s42949-026-00350-z?utm_source=chatgpt.com “
- Bressant, C. et al. (2026). Data-driven interpretability of urban tree species classification from deep learning models and Satellite Image Time Series. Urban Forestry & Urban Greening, Elsevier.
- Fatima, S. et al. (2026). Integrated deep learning framework for automating tree health assessment using ground-based images. Urban Forestry & Urban Greening, Elsevier.
- “ https://www.sciencedirect.com/science/article/pii/S1618866726000853?utm_source=chatgpt.com “
- Firoze, A. et al. (2026). Where Are the City Trees? Monitoring Urban Trees across the U.S. Using Generative AI. Communications of the ACM, 69(4), 50–59.
- “ https://dl.acm.org/doi/10.1145/3762636?utm_source=chatgpt.com “